12. Design a C program to simulate the concept of Dining-Philosophers Problem

AIM :

To design a C program to simulate the concept of Dining-Philosophers problem

CODE:

```
#include <stdio.h>
#include <stdlib.h>
#include <pthread.h>
#include <unistd.h>

#define NUM_PHILOSOPHERS 5
#define NUM_CYCLES 5

pthread_mutex_t chopsticks[NUM_PHILOSOPHERS];

void* philosopherLifeCycle(void *arg)
{
   int id = *(int *)arg;
   int left = id;
   int right = (id + 1) % NUM_PHILOSOPHERS;

   for (int i = 0; i < NUM_CYCLES; i++)
   {
      // Thinking
      printf("Philosopher %d is thinking...\n", id);
      sleep(1);
```

```
        // Deadlock prevention
        if (id == NUM_PHILOSOPHERS - 1)
        {
            pthread_mutex_lock(&chopsticks[right]);
            pthread_mutex_lock(&chopsticks[left]);
        }
        else
        {
            pthread_mutex_lock(&chopsticks[left]);
            pthread_mutex_lock(&chopsticks[right]);
        }

        // Eating
        printf("Philosopher %d is eating...\n", id);
        sleep(1);

        // Put down chopsticks
        pthread_mutex_unlock(&chopsticks[left]);
        pthread_mutex_unlock(&chopsticks[right]);
    }

    printf("Philosopher %d has finished.\n", id);
    pthread_exit(NULL);
}

int main()
{
```

```
pthread_t philosophers[NUM_PHILOSOPHERS];
int philosopher_ids[NUM_PHILOSOPHERS];

// Initialize mutexes
for (int i = 0; i < NUM_PHILOSOPHERS; i++)
{
   pthread_mutex_init(&chopsticks[i], NULL);
}

// Create philosopher threads
for (int i = 0; i < NUM_PHILOSOPHERS; i++)
{
   philosopher_ids[i] = i;
   pthread_create(&philosophers[i], NULL,
                  philosopherLifeCycle,
                  &philosopher_ids[i]);
}

// Wait for all philosophers to finish
for (int i = 0; i < NUM_PHILOSOPHERS; i++)
{
   pthread_join(philosophers[i], NULL);
}

// Destroy mutexes
for (int i = 0; i < NUM_PHILOSOPHERS; i++)
{
```

```
        pthread_mutex_destroy(&chopsticks[i]);
    }

    printf("\nAll philosophers have finished dining.\n");

    return 0;
}
```

OUTPUT:

```
Philosopher 1 is thinking...
Philosopher 3 is thinking...
Philosopher 0 is thinking...
Philosopher 2 is thinking...
Philosopher 4 is thinking...
Philosopher 1 is eating...
Philosopher 4 is eating...
Philosopher 1 is thinking...
Philosopher 4 is thinking...
Philosopher 3 is eating...
Philosopher 0 is eating...
Philosopher 3 is thinking...
Philosopher 2 is eating...
Philosopher 0 is thinking...
Philosopher 4 is eating...
```